

Solution Requirements

Date	02 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely access the application using login credentials or authenticated sessions (if login is implemented)</i>	High
2	Authorization levels	<i>Admin can manage the system and view prediction history. Users can submit applications and view only their own prediction results.</i>	High
3	External interfaces	<i>The system interacts with the trained Machine Learning model, local dataset, Flask backend, and optional SQLite database.</i>	High

4	Transactions processing	<i>User inputs are validated, preprocessed, passed to the ML model, and the prediction result is generated instantly.</i>	High
5	Reporting	<i>Display prediction results, confidence score, model accuracy, and maintain prediction history/dashboard.</i>	Medium
6	Business rules, etc.	<i>Credit card approval is determined based on applicant information such as income, employment, education, credit score, loan history, and other financial factors.</i>	High
7	Compliance to laws or regulations	<i>User data must be handled securely, respecting privacy guidelines and protecting sensitive financial information.</i>	Medium
8	Other	<i>Allow users to make multiple predictions and provide a responsive web interface for easy access.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Generate prediction results quickly after user submission.</i>	<i>Prediction response in less than 2 seconds</i>
2	Scalability	<i>The application should support increasing users and larger datasets without major performance degradation.</i>	<i>Supports 1,000+ prediction requests efficiently</i>
3	Security & Data Privacy	<i>Protect applicant information using secure coding practices and encrypted communication where applicable.</i>	<i>Secure data handling and protected user information</i>
4	Reliability & Availability	<i>The system should provide consistent prediction results with minimal downtime.</i>	<i>99% application availability</i>
5	Usability & Accessibility	<i>Provide a simple, responsive, and user-friendly interface accessible from desktop and mobile devices.</i>	<i>Easy navigation and responsive design</i>
6	Other	<i>Maintainable, modular, and well-documented code to support future updates and deployment.</i>	Clean code with proper documentation